

STATEMENT OF WORK

1. Objective/Requirements

Procure a cryocooler system designed for 100 watt heat removal at 20K temperature, including compressor, cold head, helium gas heat exchanger integrated with cold head, control panel, flex lines from compressor to cold head, and relevant documentation.

2. Characteristics, Scope, and Specs

The contractor shall perform the following duties:

- Design and construct the cryocooler system
- Ensure proper operation
- Deliver system to MSFC

The cryocooler shall have the following specifications:

- 100 watt or greater specified heat removal at 20K temperature
- Base temperature equal to or less than 17K
- Cold head shall have an integrated heat exchanger designed to cool helium gas with ½" VCR fittings for helium loop connection and a maximum allowable pressure of at least 300 psig (this is not the same specification as the compressor helium operating pressure)
- Cold head shall be mountable to a 16" OD ASA flange with 12" nozzle
- System power shall be 480VAC, 3 phase, 60 Hz
- Water cooling requirement shall be equal to or less than 5 gallons per minute at 80 F
- Flex lines from compressor to cold head shall have a length of at least 10 feet

3. Place of Performance

Vendor facility, delivered to MSFC

4. Period of Performance

MSFC shall take delivery within 9 months after award